

MONODENTATE LIGAND

Anionic

X^- - Halido
 OH^- - Hydroxo
 NO_2^- - Nitrito
 SO_4^{2-} - Sulphato
 O^{2-} - Oxo (Oxido)
 S^{2-} - Sulphido

Neutral

H_2O - Aqua
 NO - Nitrosyl
 NH_3 - Ammine
 CO - Carbonyl

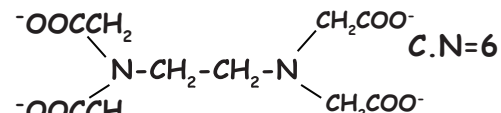
Cationic

NO_2^+ - Nitronium
 NO^+ - Nitrosonium

BIDENTATE LIGAND

- 1) Oxalato $\begin{array}{c} COO^- \\ | \\ COO^- \end{array}$
- 2) Ethane-1,2-diamine [en]
 $H_2N-CH_2-CH_2-NH_2$

POLYDENTATE LIGAND

- 1) Trien, $N(CH_2CH_2NH_2)_3$ - Tetradentate
- 2) Ethylene Diamine Tetra Acetato [EDTA]

 $C.N = 6$
 Donor - 40 Atoms
 2N Atoms

AMBIDENTATE LIGAND

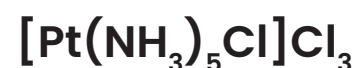
- 1) NO_2^-
- 2) SCN^-

CHELATE LIGAND

- 1) $EDTA^{4-}$, en, ox
- 2) greater Stability

WERNER'S THEORY

- 1) 1^o Valency - O.N
- 2) 2^o Valency - C.N



- 1) 1^o Valency - O.N = 4
- 2) 2^o Valency - C.N = 6
- 3) AgCl Formed per mole of complex $\rightarrow 3$
- 4) Total ions per mole of complex - 4

GEOMETRICAL ISOMERISM

- 1) $[Ma_2b_2]$ - 2 (cis + trans)
- 2) $[Ma_2bc]$ - 2 (cis + trans)
- 3) $[Mabcd]$ - 3
- 4) $[Ma_4b_2]$ / $[Ma_4bc]$ - 2 (cis + trans)
- 5) $[Ma_3b_3]$ - fac & mer
- 6) $[Ma_2b_2c_2]$ - 5
- 7) $[Mabcdef]$ - 15
- 8) $[M(en)_2b_2]$ or $[M(en)_2bc]$ - 2

OPTICAL ISOMERISM



2 Optical isomers (d cis + l cis)

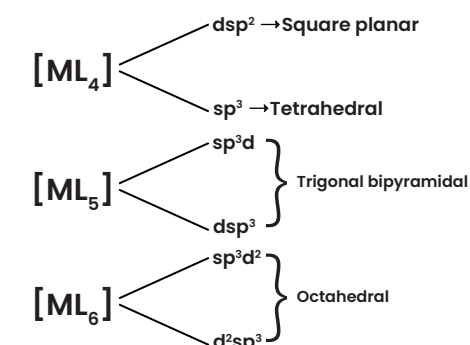


Total 3 Stereoisomer (d cis + l cis + trans)

VBT

SPECTROCHEMICAL SERIES

$I^- < Br^- < SCN^- < Cl^- < F^- < C_2O_4^{2-} < H_2O$
 $< NCS^- < EDTA^{4-} < NH_3 < en < NO_2^- < CN^- < CO$



COORDINATION COMPOUNDS



UNPAIRED ELECTRONS - PARAMAGNETIC
 PAIRED ELECTRONS - DIAMAGNETIC

$$\mu = \sqrt{n(n+2)}$$

1=1.73 BM
 2=2.84 BM
 3=3.87 BM
 4=4.90 BM
 5=5.92 BM

$[NiCl_4]^{2-}$ - sp^3 , Tetrahedral $n=2$
 $[NiCO_4]$ - sp^3 , Tetrahedral $n=0$
 $[Ni(CN)_4]^{2-}$ - dsp^2 , Sq. planar $n=0$

H_2O act as SFL with Co^{3+}

Cr^{3+} - $\uparrow\uparrow\uparrow\uparrow$ forms d^2sp^3

4d and 5d metals with all ligands act as SFL

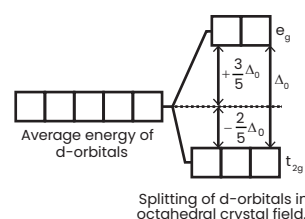
$[Pt(CN)_4]^{2-}$, $[PtCl_4]^{2-}$, $[PdCl_4]^{2-}$ = dsp^2 (diamagnetic)

$C_2O_4^{2-}$ act as SFL with Co^{3+}

F^- act as SFL with Ni^{4+}

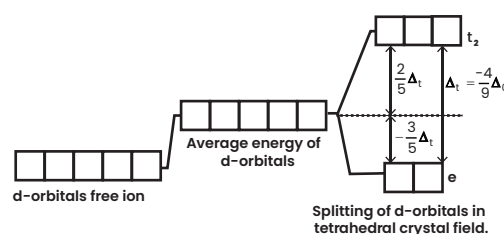
CFT

- 1) Octahedral - Ligand approaches along the axis ($e_g > t_{2g}$)



$$CFSE = a \times (-0.4 \Delta_o) + b \times (0.6 \Delta_o) + np$$

- 2) Tetrahedral - Ligand approaches b/w the axis ($t_2 > e$)



$$CFSE = a \times (-0.6 \Delta_t) + b \times (0.4 \Delta_t) + np$$

here a, b are number of electrons in low energy and high energy d-orbitals

HIGH SPIN COMPLEX
 $\Delta_o < \text{Pairing Energy}$

LOW SPIN COMPLEX
 $\Delta_o > \text{Pairing Energy}$

For calculation of n, Crystal field stabilisation energy consider newly formed pairing only

CFSE

- 1) Stability $\propto CFSE$
- 2) $\Delta_o > \Delta_t$, $\Delta_t = -\frac{4}{9} \Delta_o$
- 3) $[Fe(CN)_6]^{4-} < [Fe(CO)_6]$
- 4) Jahn Teller distortion by - d^4 , d^7 , d^9
- 5) Dissociation constant = $\frac{1}{\beta_n}$ (β_n = stability constant)
- 6) Color of d-d transition, $E \propto \frac{1}{\lambda}$
 $[Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+} < [Co(H_2O)_6]^{3+}$
 $E \uparrow \lambda \downarrow$ (E = energy of d-d transition) $E \downarrow \lambda \uparrow$

OMC

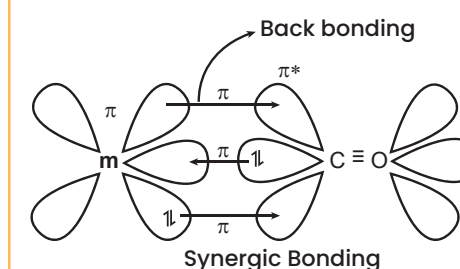
- 1) σ bonded - $RMgX$ (Grignard reagent) - R_2SiCl_2

- 2) π bonded - Ferrocene

- 3) σ & π bonded - Metal Carbonyls Mononuclear - $[Fe(CO)_6]$

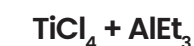


B.O. $\uparrow$ B.L. $\downarrow$ (w.r.to C-O bond) B.O. $\downarrow$ B.L. $\uparrow$



APPLICATION OF ORGANO METALLIC COMPOUNDS

- 1) Ziegler Natta Catalyst is used in polymerisation of alkene



- 2) Wilkinson's Catalyst is used in hydrogenation



APPLICATION OF COORDINATION COMPOUNDS

- 1) Cisplatin ($[Pt(NH_3)_2Cl_2]$) is used as an anticancerous agent
- 2) EDTA is used in Lead poisoning
- 3) Copper and iron poisoning - penicillamine & deferoxamine
- 4) Vitamin B-12 - Cobalt
- 5) Chlorophyll - Magnesium
- 6) Carboxy peptidase - Zinc